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Review article

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Optimizing the safety of antibody–drug conjugates for patients with solid tumours

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Abstract

Over the past 5 years, improvements in the design of antibody–drug conjugates (ADCs) have enabled major advances that have reshaped the treatment of several advanced-stage solid tumours. Considering the intended rationale behind the design of ADCs, which is to achieve targeted delivery of cytotoxic molecules by linking them to antibodies targeting tumour-specific antigens, ADCs would be expected to be less toxic than conventional chemotherapy. However, most ADCs are still burdened by off-target toxicities that resemble those of the cytotoxic payload as well as on-target toxicities and other poorly understood and potentially life-threatening adverse effects. Given the rapid expansion in the clinical indications of ADCs, including use in curative settings and various combinations, extensive efforts are ongoing to improve their safety. Approaches currently being pursued include clinical trials optimizing the dose and treatment schedule, modifications of each ADC component, identification of predictive biomarkers for toxicities, and the development of innovative diagnostic tools. In this Review, we describe the determinants of the toxicities of ADCs in patients with solid tumours, highlighting key strategies that are expected to improve tolerability and enable improvements in the treatment outcomes of patients with advanced-stage and those with early stage cancers in the years to come.

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Key points

- Indications for the use of antibody–drug conjugates (ADCs) are rapidly expanding, with development progressively moving from the advanced-stage to the early stage setting, and from monotherapy to combination strategies.
- Despite being designed with the rationale of expanding the therapeutic indices of conventional chemotherapies, most ADCs have a toxicity profile similar to that of their cytotoxic payload.
- Unconventional and potentially life-threatening toxicities can also be observed with certain ADCs, requiring an increased understanding of these events and the optimization of diagnostic and management practices.
- Multiple pharmacological modification strategies are being pursued in an attempt to improve the tolerability of ADCs, including molecular alterations of the antibody moiety, linker and/or cytotoxic payload.
- Exploration of different doses within randomized trials and investigation of response-adapted dosing strategies could enable optimized use of ADCs, which could maximize their therapeutic value for each indication.
- Extensive efforts to identify biomarkers of toxicities in patients receiving ADCs and to develop diagnostic tools enabling the anticipation and/or early detection of toxicities are currently ongoing.

Introduction

The targeted delivery of cytotoxic payloads through antibody–drug conjugates (ADCs) is rapidly emerging as a highly effective strategy to treat multiple types of cancer[1]. A total of 12 ADCs are currently approved by the FDA and/or EMA for patients with various cancers[2,3], with 6 of these approved for solid tumour indications. Importantly, most of these approvals have occurred in the past 4 years, reflecting an expanding pipeline of promising agents that is expected to lead to further approvals in the near future[4]. Together with improving outcomes for patients with a range of tumour types, ADCs are challenging the current approach to the categorization of cancer: HER2-targeted ADCs have shown activity in breast and gastric cancers deemed to be HER2-negative by traditional criteria[5–10], leading to the definition of a novel 'HER2-low' subset of targetable tumours[11]. Although only preliminary data are currently available, this novel subset might expand further across tumour types based on positive results from the DESTINY-Pantumor02 phase II trial[12]. The cross-histological activity of ADCs and the expansion of their reach beyond canonical biomarker-defined subtypes have a crucial repercussion: a large and rapidly increasing proportion of patients with cancer are becoming eligible for treatment with ADCs. This is particularly true for patients with advanced-stage cancers but also includes some patients with early stage solid tumours, a setting in which multiple trials testing ADCs are currently ongoing in an attempt to prevent disease recurrence and thus improve cure rates.

ADCs were originally designed with the aim of improving the therapeutic index of conventional chemotherapies by enabling the selective delivery of cytotoxic payloads to cancer cells expressing a tumour-associated antigen targeted by the antibody component of the conjugate[13]. However, this objective has not been clearly achieved by the current generation of ADCs: indeed, after normalizing for cytotoxin content, ADCs have been found to have a similar maximum-tolerated dose to that of their unconjugated cytotoxic payload, and have not yet been proven to lower the minimum effective dose required for antitumour activity. Rather, data suggest that, when dosed at or near their maximum-tolerated doses, certain ADCs have enhanced antitumour activity compared to that of the closely related unconjugated cytotoxic compound[14]. In numerical terms, a meta-analysis of data from 169 clinical trials testing ADCs found that these agents are associated with treatment-related adverse events in >90% of patients (including grade ≥3 events in 40%), with extensive variability in the types of toxicities and their incidence depending on the specific composition of the conjugate[15]. Furthermore, safety concerns have led to discontinuation of the clinical development of numerous initially promising ADCs. For example, the delta-like protein 3 (DLL3)-targeted ADC rovalpituzumab tesirine, which reached phase III testing in patients with small-cell lung cancer, was found to be less tolerable and to provide inferior overall survival compared to topotecan[16], leading to discontinuation of further development. Similarly, the EGFR-targeted construct depatuxizumab mafodotin[17] was discontinued after the emergence of relevant toxicities plus no improvement in overall survival when added to standard-of-care therapy in a phase III trial involving patients with glioblastoma.

Given the large proportion of patients with cancer currently receiving ADCs as well as the expected expansion in the use of these compounds to patients with early stage disease, improving our understanding of the adverse effects of ADCs is becoming increasingly important to enable safe implementation in clinical practice and to optimize the design of novel conjugates. In this Review, we discuss current knowledge of the toxicities of ADCs in patients with solid tumours, highlighting ongoing efforts to optimize the safety of these agents in clinical practice.

ADCs as targeted chemotherapy

Despite extensive differences in structure, all approved ADCs share three key elements: an engineered monoclonal antibody targeting a tumour-associated antigen, a cytotoxic payload and a molecular linker connecting the payload with the antibody[13]. Such a structure has the rationale of merging the selectivity of an antibody with the antitumour activity of cytotoxic molecules with the aim of delivering the highest possible dose of the cytotoxic drug to the tumour while also sparing non-malignant tissues. In addition to delivering a cytotoxic payload, ADCs have been shown to retain the antitumour effects of the antibodies, including inhibition of oncogenic signalling pathways and enhancement of antitumour immunity through the induction of antibody-dependent cellular cytotoxicity[18]. In this framework, ADCs can neither be solely intended as chemotherapies nor as targeted therapies; they rather represent a complex combination of these two treatment modalities, often considered to be 'targeted chemotherapies'. This combination is reflected in both the antitumour activity and toxicities of this class of agents (Fig. 1 and Table 1).

Given their large molecular size, ADCs are generally administered via intravenous injection, either as a bolus dose or an infusion; subcutaneous administration of ADCs has not yet been proven to be safe, and data from in vivo experiments suggest that this route of administration could be associated with reduced activity and severe skin toxicities for some ADCs[19]. After injection, ADCs are ideally meant to remain intact while in circulation and to only release their cytotoxic payload within or

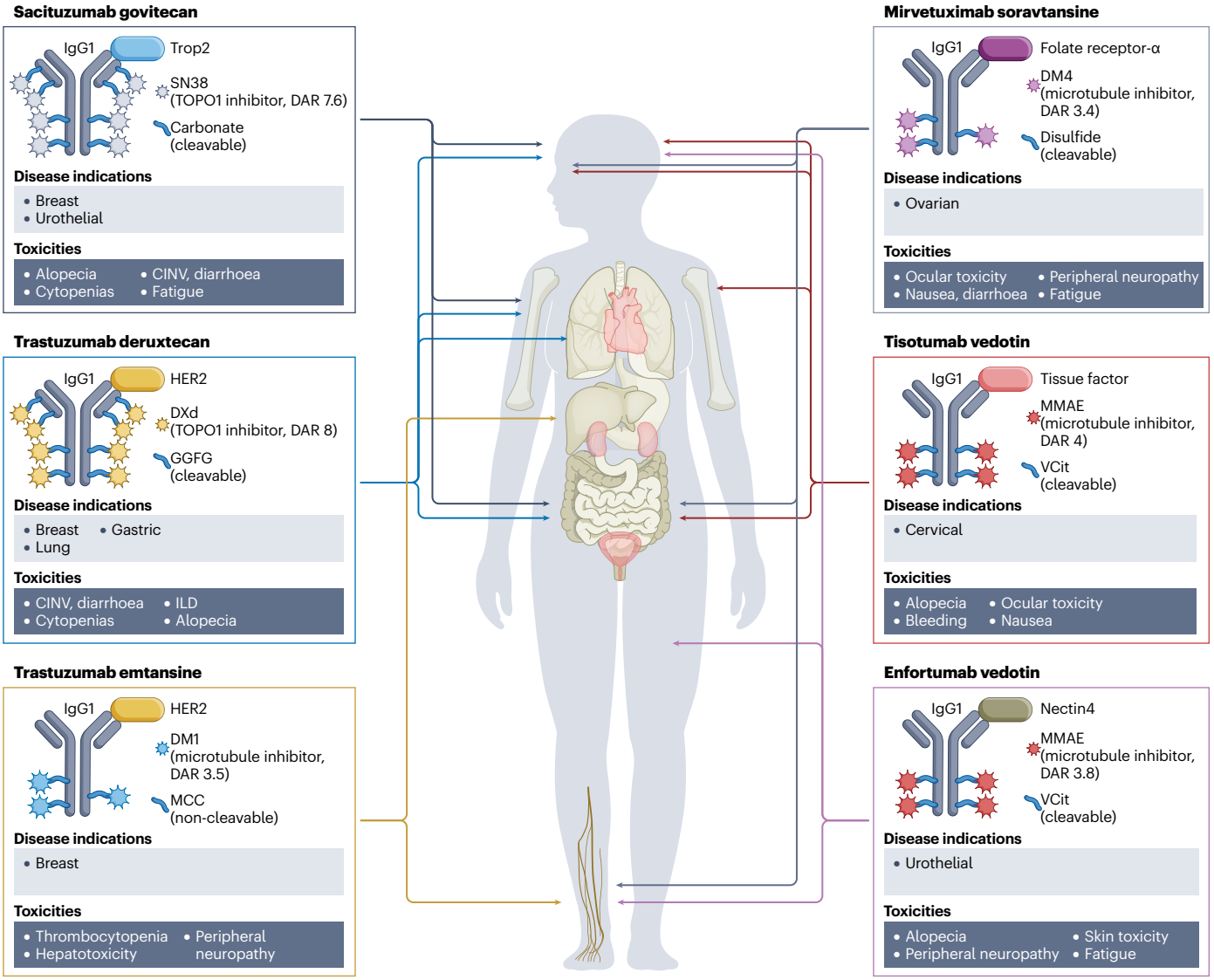


Fig. 1 | Structure and main toxicities of ADCs currently approved for patients with solid tumours. Thus far, six antibody–drug conjugates (ADCs) have been approved by the FDA and/or EMA for patients with solid tumours. This figure depicts the composition of each of these ADCs (in terms of targeted antigen, type of monoclonal antibody, payload and linker), the currently approved indications and the most common toxicities observed with each agent. The adverse effect profile of each ADC is often a mixture of on-target and off-target effects, with the latter generally determining the maximum-tolerated dose. Common adverse effects seen to varying degrees across many ADCs include fatigue, alopecia, cytopenias and gastrointestinal disturbances, although individual ADCs can also be associated with more unique toxicities related to the stability of the compound and the complex interactions of various different components of the conjugate. CINV, chemotherapy-induced nausea and vomiting; GGFG, Gly-Gly-Phe-Gly; DAR, drug-to-antibody ratio; DXd, deruxtecan; ILD, interstitial lung disease; MCC, maleimidomethyl cyclohexane-1-carboxylate; MMAE, monomethyl auristatin E; TOPO1, topoisomerase 1; Trop2, trophoblast cell-surface antigen 2; VCit, valine-citrulline.

in close proximity to the targeted tumour cells. Each ADC characteristic has a crucial role in achieving this objective, with small modifications in structure often leading to dramatic changes in the pharmacokinetic and/or pharmacodynamic profile of the agent (Table 2). After injection, ADCs typically circulate in the bloodstream as a dynamic mixture of the intact ADC (>90% of the composition)[20], the unconjugated drug (or drug–linker) and the dissociated antibody[13]. From the circulation, ADCs progressively diffuse into the interstitial spaces of body tissues, ultimately reaching the targeted tumour cells at a fraction estimated to be approximately 0.1% for solid tumours[21,22]. The limited fraction of ADCs reaching tumour cells underscores the need to utilize potent cytotoxic molecules, which, with few exceptions, are generally unsuitable for unconjugated administration. Once an ADC reaches the tumour microenvironment (TME), it is generally thought to bind the target antigen expressed on the cancer cell surface, undergo internalization within endosomes, and release the payload via chemical or enzymatic cleavage in the lysosomes, ultimately leading to either necrosis or apoptosis depending on the mechanism of action of the payload and the concentration reached in the target site[23]. More hydrophobic payloads (such as monomethyl auristatin E (MMAE) and exatecan derivatives) are able to

diffuse outside of the target cells following intracellular deconjugation from the antibody, thus exerting a 'bystander killing' effect on antigen-negative cells that is thought to enhance the antitumour activity of certain ADCs[24–26]. This effect can be associated with increased efficacy and killing of cancer cells in tumours with limited or heterogeneous expression of the target antigen[27], although it also constitutes a determinant of toxicity[28]: the released payload can also enter adjacent non-malignant cells via passive diffusion or transporter-mediated uptake, potentially leading to off-target cytotoxicity[28]. Beyond the traditionally accepted mechanism of targeted payload release within tumour cells,

Table 1 | Toxicities of FDA-approved ADCs in patients with solid tumours

ADC	Indication (year of FDA approval)	ADC characteristics	Most common AEs (any grade)	Most common grade ≥3 AEs
Trastuzumab emtansine	HER2-positive unresectable or metastatic breast cancer (2013)	DM1 (a maytansinoid) conjugated to an anti-HER2 IgG1 antibody via a non-cleavable MCC thioether linker	Nausea (39%), fatigue (35%), thrombocytopenia (28%), elevated AST (22%), vomiting (19%), elevated ALT (17%), anaemia (10%)[42]	Thrombocytopenia (13%), elevated AST (4%), elevated ALT (3%), anaemia (3%), fatigue (2%), hypokalaemia (2%), neutropenia (2%), diarrhoea (2%)[42]
	HER2-positive early stage breast cancer with residual disease following neoadjuvant treatment (2019)		Fatigue (50%), nausea (42%), decreased platelet count (29%), elevated AST (28%), headache (28%), arthralgia (26%), radiation skin injury (25%), elevated ALT (23%)[169]	Decreased platelet count (6%), hypertension (2%), radiation-related skin injury (1%), peripheral sensory neuropathy (1%), decreased neutrophil count (1%), hypokalaemia (1%), fatigue (1%), anaemia (1%)[169]
Sacituzumab govitecan	Metastatic TNBC or HR-positive, HER2-negative breast cancer (2021, 2023)	SN-38 (a camptothecin) conjugated to an anti-Trop2 IgG1 antibody via a hydrolysable carbonate linker	Neutropenia (63%), diarrhoea (59%), nausea (57%), alopecia (46%), fatigue (45%), vomiting (29%), nervous system disorders (25%), reduced appetite (20%)[45,188]	Neutropenia (51%), diarrhoea (10%), leukopenia (10%), anaemia (8%), febrile neutropenia (6%), fatigue (3%), nausea (2%), infection (2%)[45,188]
	Locally advanced or metastatic urothelial carcinoma (2021)[a]		Diarrhoea (65%), nausea (60%), fatigue (52%), alopecia (47%), neutropenia (46%), decreased appetite (36%), anaemia (33%), vomiting (30%), leukopenia (25%)[189]	Neutropenia (35%), leukopenia (18%), anaemia (14%), diarrhoea (10%), febrile neutropenia (10%), lymphopenia (7%), urinary tract infection (6%)[189]
Trastuzumab deruxtecan	HER2-positive unresectable or metastatic breast cancer (2019)	DXd (a camptothecin) conjugated to an anti-HER2 IgG1 antibody via a protease-cleavable tetrapeptide (GGFG) linker	Nausea (73%), fatigue (45%), vomiting (44%), neutropenia (43%), alopecia (36%), anaemia (30%), leukopenia (30%), reduced appetite (26%), pneumonitis/ILD (11%)[190]	Neutropenia (19%), thrombocytopenia (7%), leukopenia (7%), nausea (7%), anaemia (6%), fatigue (5%), vomiting (2%), elevated ALT (2%), pneumonitis/ILD (1%)[190]
	HER2-low unresectable and/or metastatic breast cancer (2022)		Nausea (73%), fatigue (48%), alopecia (38%), vomiting (34%), neutropenia (33%), anaemia (33%), reduced appetite (29%), pneumonitis/ILD (10%)[191]	Neutropenia (14%), anaemia (8%), fatigue (8%), leukopenia (7%), nausea (5%), elevated AST (3%), reduced appetite (2%), pneumonitis/ILD (1%)[b,191]
	Metastatic *HER2*-mutant NSCLC (2022)		Any-grade AEs (92%), any-grade ILD (6%)[c,129]	Grade ≥3 AEs (32%), grade ≥3 ILD (1%)[c,129]
	Unresectable and/or metastatic HER2-positive gastric cancer (2021)		Nausea (63%), neutropenia (63%), reduced appetite (60%), anaemia (58%), reduced platelet count (39%), leukopenia (38%), malaise (34%), diarrhoea (32%), pneumonitis/ILD (9%)[192]	Neutropenia (51%), anaemia (38%), leukopenia (21%), reduced appetite (17%), lymphopenia (11%), reduced platelet count (10%), fatigue (7%), nausea (5%), pneumonitis/ILD (2%)[192]
Enfortumab vedotin	Locally advanced or metastatic urothelial carcinoma (2021)	MMAE (an auristatin) conjugated to an anti-nectin 4 IgG1 antibody via a protease-cleavable maleimidocaproyl valine-citrulline linker	Alopecia (45%), peripheral sensory neuropathy (34%), pruritus (32%), fatigue (31%), diarrhoea (24%), dysgeusia (24%), nausea (23%), maculopapular rash (16%)[40]	Maculopapular rash (7%), fatigue (6%), neutropenia (5%), diarrhoea (3%), peripheral sensory neuropathy (3%), reduced appetite (3%), anaemia (1%), leukopenia (1%)[40]
Mirvetuximab soravtansine	Platinum-resistant ovarian cancer (2022)[a]	DM4 (a maytansinoid) conjugated to an anti- FRα IgG1 antibody via a cleavable sulfo-SPDB disulfide linker	Blurred vision (41%), nausea (29%), keratopathy (29%), dry eye (25%), fatigue (24%), diarrhoea (22%)[114]	Keratopathy (9%), blurred vision (6%), dry eye (2%), neutropenia (2%), diarrhoea (2%), fatigue (1%)[114]
Tisotumab vedotin	Recurrent or metastatic cervical cancer (2021)[a]	MMAE (an auristatin) conjugated to an anti-tissue factor IgG1 antibody via a protease-cleavable maleimidocaproyl valine-citrulline linker	Alopecia (38%), epistaxis (30%), nausea (27%), conjunctivitis (26%), fatigue (24%), dry eye (23%)[64]	Neutropenia (3%), fatigue (2%), peripheral sensory neuropathy (2%), ulcerative keratitis (2%), anaemia (1%)[64]

ADC, antibody–drug conjugate; AE, adverse event; ALT, alanine aminotransferase; AST, aspartate aminotransferase; DXd, deruxtecan; FRα, folate receptor-α; GGFG, Gly-Gly-Phe-Gly; HR, hormone receptor; ILD, interstitial lung disease; MCC, maleimidomethyl cyclohexane-1-carboxylate; MMAE, monomethyl auristatin E; NSCLC, non-small-cell lung cancer; TNBC, triple-negative breast cancer; Trop2, trophoblast cell-surface antigen 2. [a]Accelerated approval only. [b]Includes three treatment-related deaths. [c]Data available from meeting abstract only.

Table 2 | Pharmacological determinants of the toxicity of ADCs

Category	Mechanism	Refs.
Pharmacodynamic		
Uptake of intact ADCs into non-malignant cells	Uptake by Fc and/or C-type lectin receptors	24,31
	Non-specific endocytosis	31
	Endocytosis following on-target, off-tumour activity	31
	Passive diffusion	28
Bystander effect	Payload released from apoptotic cancer cells	25
	Payload released from apoptotic non-malignant cells	25
Deconjugation and linker stability	Premature release of payload in the systemic circulation	14
Drug-to-drug antibody ratio	Increased total payload dose	56
Tumour penetration and accumulation	Limited tumour penetration increases systemic toxicity	131
Pharmacokinetic		
Target-specific binding	Limited target-specific binding increases non-target exposure	68,69
Absorption and distribution	Increased systemic concentration	32
Decreased catabolism	Increased systemic concentration	32
Excretion properties	Prolonged exposure	32
Increased half-life	Prolonged exposure	32
Proteolysis-mediated catabolism	Prolonged exposure	32
Albumin binding	Prolonged exposure	14,33
Payload biochemical properties	Lipophilic payloads have higher tissue penetrance	14,34,48

ADC, antibody–drug conjugate.

certain ADCs do not require antigen engagement and internalization in order to release the cytotoxic payload[29]. For example, sacituzumab govitecan was shown to release its SN-38 payload extracellularly within the TME, and this mechanism possibly explains both the activity and toxicities seen with this and other ADCs[30].

As mentioned above, most parts of the ADC do not reach tumour cells and are progressively degraded and then eliminated via a combination of specific and non-specific mechanisms, including target-mediated clearance, Fcγ receptor (FcγR)-mediated uptake and/or pinocytosis by macrophages located in a variety of tissues[31,32]. Importantly, before undergoing degradation and excretion, payload–linker complexes released from ADCs utilizing thiol-maleimide chemistries (accounting for most of the approved ADCs) can react with the free cysteine residues of serum albumin, thus producing long-lived albumin–linker–payload adducts[14,33]. Although still incompletely characterized, this phenomenon might have implications for both the activity and toxicity profile of certain ADCs via albumin-conjugate tumour uptake and non-specific disposition in non-malignant tissues that ultimately prolongs the half-life of the cytotoxic payload after each infusion[14,34].

Dissecting the toxicities of ADCs

ADCs are modular agents and small modifications of any of their key components can lead to major changes in clinical profile[35–38] (Fig. 2). In this section, we dissect the relative contributions of each ADC component and the role of patient characteristics in determining the type and severity of the toxicities observed in patients receiving these agents.

Payload

On the basis of the rationale behind the development of ADCs, the identity of the targeted antigen would be expected to determine the toxicity profile of the drug. However, clinical experience indicates that most of the adverse events associated with ADCs remain similar in spectrum, incidence and severity to those of the payload backbone[14,15], and different ADCs that share the same payload often have similar toxicity profiles regardless of differences in the target antigen[39]. These toxicities can be broadly divided into off-target, off-tumour effects that are unrelated to the antigen targeted by the antibody vehicle, and on-target, off-tumour effects resulting from engagement of the antibody with cognate antigen located in non-malignant tissues.

Off-target, off-tumour toxicities dominate the toxicity profile of most ADCs[15], often resulting in an adverse effect profile resembling that of the backbone payload[14,15]. For example, ADCs carrying auristatins (synthetic analogues of dolastatin with antimitotic activity relating to F-actin stabilization) commonly lead to peripheral neuropathies. For example, any-grade peripheral neuropathies have been reported in 46.3% of patients with advanced-stage urothelial carcinoma receiving enfortumab vedotin[40], which is similar to the incidence of such events in patients receiving unconjugated dolastatin (any-grade peripheral neuropathies have been reported in 36% of patients with solid tumours receiving dolastatin 10)[41]. ADCs carrying an antimitotic microtubule-targeting maytansine derivative (DM1) payload can induce hepatotoxicity and thrombocytopenia (any-grade forms of both events occur in approximately 30% of patients with breast cancer receiving trastuzumab ematansine (T-DM1))[42] similar to unconjugated maytansine (any-grade hepatotoxicity in up to 43% of patients with solid tumours and thrombocytopenia in up to 10%)[43,44]. ADCs with camptothecin-class topoisomerase I inhibitor payloads (such as SN-38, belotecan and deruxtecan (DXd)) commonly cause alopecia, diarrhoea and neutropenia (any-grade forms of these events have been observed in >50% of patients with triple-negative breast cancer (TNBC))[45,46], similar to irinotecan (any-grade forms of these events reported in >50% of patients with colorectal cancer)[47]. The key mechanism of off-target, off-tumour toxicities with the currently approved ADCs is thought to, at least in part, relate to the premature deconjugation of the payload within the systemic circulation, which leads to the diffusion of free cytotoxic payload into extratumoural compartments[14]. Such payloads are generally lipophilic molecules, which are able to permeate plasma membranes and enter non-target, non-malignant cells[48]. As previously mentioned, part of the payload can also become bound to serum albumin and other circulating thiol-containing plasma proteins, which can increase the half-life of the payload–linker complex and potentially lead to deposition of the payload in non-malignant tissues[14,34]. Beyond payload detachment from the ADC, other mechanisms have been suggested to mediate the exposure of non-malignant cells to the cytotoxic payload, including non-specific endocytosis of the intact ADC within non-malignant cells[24] as well as off-target, receptor-mediated uptake resulting from the interaction of the Fc domain of the antibody backbone with Fc receptors expressed by immune cells[24]. These latter mechanisms might be more relevant to highly stable ADCs, which are associated with limited premature deconjugation of the payload and release into the circulation and are thus more likely to

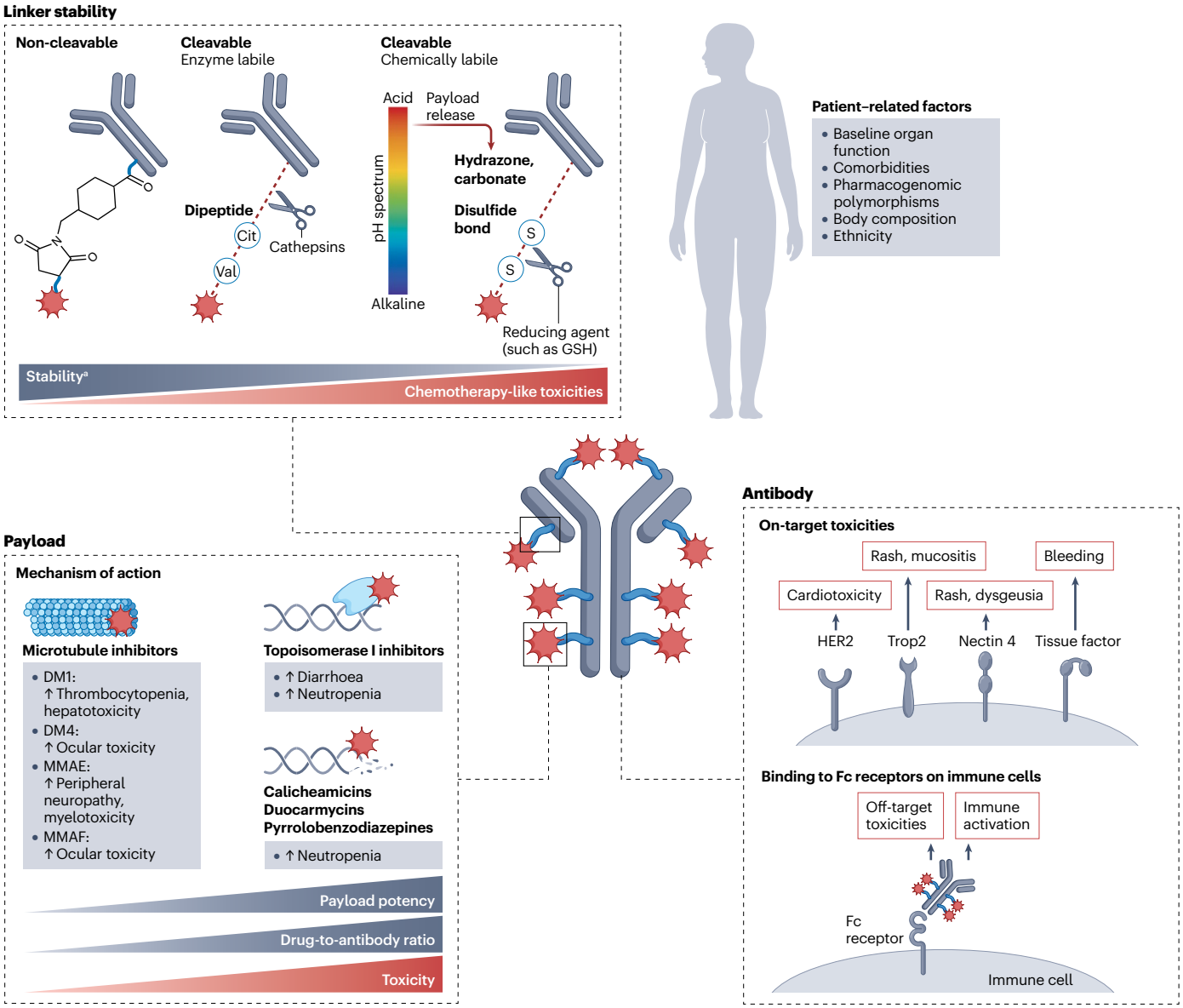


Fig. 2 | Determinants of the toxicities of ADCs. The adverse effect profile of antibody–drug conjugates (ADCs) is often dominated by off-target, off-tumour toxicities that usually resemble those of the cytotoxic payload carried by the conjugate. Therefore, payload selection can have major implications for the expected toxicity profile of ADCs. However, the type, incidence and severity of these toxicities are also a function of the stability of the linker used to bind the payload to the antibody, with a less-stable linker expected to lead to increased chemotherapy-related toxicities owing to higher peak concentrations of released payload, and a more-stable linker sometimes associated with an increased incidence of unexpected adverse effects such as ocular toxicities. The antibody component of ADCs can contribute to the toxicity profile through the induction of on-target, off-tumour toxicities as well as via engagement of Fcγ receptors on immune cells. Lastly, extensive interpatient variability in the toxicity of the same ADC can be observed depending on multiple patient-related factors, including baseline organ function, the presence of comorbidities, pharmacogenomic polymorphisms, body composition and ethnicity. Cit, citrulline; GSH, glutathione; MMAE, monomethyl auristatin E; MMAF, monomethyl auristatin F; Trop2, trophoblast cell-surface antigen 2; Val, valine. [a]Highly stable linkers have been associated with an increase in the incidence of certain adverse effects, including ocular toxicities, neurotoxicities or hepatotoxicities, depending on the specific agent.

encounter non-malignant tissues as intact ADCs[49]. Regardless of the mechanism, the extent of off-target exposure of non-malignant cells to the payload ultimately determines the tolerability of the agent, making the choice of the payload a crucial decision in the design of any ADC.

Linker

As mentioned above, the main mechanism leading to off-target toxicities in patients receiving ADCs probably relates to the timing and localization of the release of the payload from the conjugate. These features are largely determined by the stability and pharmacological

structure of the linker, which can therefore have major implications for the toxicity profile of the ADC[49,50].

An ideal linker is sufficiently stable to deliver the payload to the intended site, yet also labile enough to release an effective amount of payload either within or in close proximity to the tumour. Non-cleavable linkers, such as maleimidomethyl cyclohexane-1-carboxylate (as utilized in T-DM1), require intracellular degradation of the ADC to release the payload, resulting in stability in the circulation (relative to ADCs with cleavable linkers). Following antigen binding and intracellular release of the payload, the presence of charged lysines or cysteines limits the cell membrane permeability of the unconjugated payload and renders it unlikely to re-enter the circulation or diffuse into bystander cells[35]. Conversely, cleavable linkers have chemically or enzymatically labile chemical structures, which can lead to payload release through several mechanisms, including acidic degradation (hydrazones and carbonates), cleavage by plasma and lysosomal proteases (peptides and glucuronides), or thiol-disulfide exchange reactions (disulfides)[51]. In general, less-stable linkers lead to earlier release of free payload into the circulation, with a higher peak cytotoxin concentration and an increase in typical chemotherapy-related toxicities (such as cytopenias, alopecia and/or gastrointestinal toxicities)[52]. However, more-stable linkers can lead to prolonged circulation of the intact ADC, with the payload released later and in variable body locations beyond the circulation[14,52]. This aspect might account for the peculiar toxicity profiles of certain highly stable ADCs, several of which have been found to have limited chemotherapy-related toxicities but unexpectedly high incidences of ocular toxicities[49,53–55]. These findings suggest that a balance should be pursued when attempting to determine ADC stability, and that excessive payload release and payload retention by the ADC can both lead to unintended toxicities.

Beyond linker stability, both the specific chemistry used to link payloads with antibodies and the drug-to-antibody ratio (DAR) can have implications for the toxicity profile of ADCs. Stochastic conjugation, namely the non-selective linking of payloads to lysine or cysteine residues located in the antibody, produces a mixture of ADCs with heterogenous DARs[56], which can lead to variations in the toxicity of the conjugate[50]. The manufacture of T-DM1 involves stochastic lysine bioconjugation, which generates a heterogenous admixture of conjugates with an average DAR of 3.5, with the payload located at variable conjugation sites around the antibody[35]. The manufacturing processes for trastuzumab deruxtecan (T-DXd) and sacituzumab govitecan instead involve stochastic cysteine conjugation, which enables drug conjugation at eight different regions of the antibody, enabling a DAR of up to 8, with reduced heterogeneity compared with lysine conjugation[50]. All other approved ADCs, including those for haematological indications, are produced using stochastic conjugation methods involving either lysine or cysteine[50]. Strategies designed to enable more consistent payload conjugation are an area of active development and will be further discussed later in the Review.

Antibody

Much of the toxicity of ADCs is determined by the linker–payload complex[14], although even ADCs with the same payload and linker can differ substantially in their adverse effect profile owing to on-target, off-tumour toxicities. These events are related to the engagement of a specific target or the accumulation of payload in non-malignant tissues expressing the ADC target and thus vary broadly depending on the target antigen. This phenomenon was initially observed with the LeY-targeted conjugate BR96–doxorubicin, which was associated with gastrointestinal tract toxicities (where LeY is highly expressed) and less frequently with the haematological or cardiac toxicities typically seen with unconjugated doxorubicin[57]. Cardiotoxicities can be observed with T-DM1 or T-DXd and are largely related to the on-target, off-tumour effects of trastuzumab[58] as reflected by the observation that such toxicities are less frequent with other DXd-based ADCs. Instead, the trophoblast cell-surface antigen 2 (Trop2)-targeted ADC datopotamab deruxtecan (Dato-DXd) is associated with high incidences of rash and stomatitis[59], similar to other Trop2-targeted ADCs (such as PF-06664178)[60], possibly owing to high levels of Trop2 expression in skin and mucosal tissues[61]. The HER3-targeted ADC patritumab deruxtecan (HER3-DXd) is associated with a high risk of thrombocytopenia[62], an adverse effect that has also been observed with unconjugated patritumab[63]. Similarly, despite many MMAE (also known as vedotin)-based ADCs having similar toxicity profiles (typically characterized by anaemia, neutropenia and peripheral neuropathy)[40,64,65], which are mostly related to their payload, more unique adverse effects can be observed depending on the targeted antigen. For example, the anti-nectin 4 ADC enfortumab vedotin is associated with severe skin toxicities and dysgeusia[40], probably owing to high levels of nectin 4 expression in the skin and salivary glands[66], whereas treatment with tisotumab vedotin is associated with a substantial risk of bleeding[64], which might be related to targeting tissue factor (also known as coagulation factor III)[67]. In certain scenarios, on-target toxicities might dominate the safety profile of the ADC: this is the case for the EphA2-targeted MMAF-based ADC MEDI-547, which is associated with life-threatening bleeding and coagulation events, even at low doses, that are potentially related to the on-target engagement of EphA2, a receptor implicated in neoangiogenesis[68,69]. To minimize the risk of such on-target toxicities, a careful selection of ADC targets is warranted with preference recommended for antigens with differential levels of expression between tumour cells (ideally high expression) and non-malignant cells (ideally low or absent expression)[70].

Beyond inducing on-target toxicities through their antigen-binding domains, the antibody component of the ADC can also lead to toxicities through interactions of their Fc domain with FcγRs expressed on immune cells or other non-malignant cells[24]. This mechanism has been suggested to be implicated in the induction of interstitial lung disease (ILD) by deruxtecan-based ADCs owing to the uptake of these conjugates by alveolar macrophages[71], which express high levels of FcγR[72,73]. Induction of ILD has indeed been observed, with variable incidences, with several ADCs seemingly regardless of the targeted antigen (including HER2, HER3 and Trop2)[74] and has not been observed in monkeys exposed to high doses of unconjugated DXd[71], suggesting that this effect might not be related to the targeting of any specific epitope nor to the payload alone. Similarly, engagement of the antibody Fc domain with FcγRs expressed on megakaryocytes might have a role in T-DM1-induced thrombocytopenia[75], although controversies remain regarding this mechanism[76].

Overall, although antibody-related toxicities do not generally dominate the adverse effect profile of ADCs, several instances of severe on-target, off-tumour toxicities or toxicities mediated by the Fc portion of the antibody have been observed. These observations highlight the complexity of the mechanism of action of these compounds and the relevance of each ADC component to the tolerability profile.

Patient-related factors

Beyond the differences in toxicity profiles observed among various ADCs, a relevant degree of heterogeneity also exists in the spectrum

and grade of adverse effects occurring in different patients who receive the same ADC. Multiple patient-related factors can affect the pharmacokinetics and pharmacodynamics of these agents, including baseline organ function, the presence of comorbidities, polymorphisms in enzymes involved in the metabolism of ADCs or their catabolites, and the body composition of each patient.

Impaired baseline organ function and/or specific comorbidities can affect both the metabolism of the ADC and the level of susceptibility to further organ damage. Impaired baseline renal function and lower baseline oxygen saturation have both been associated with a higher risk of pulmonary toxicities from T-DXd[77], whereas impaired liver function has been found to increase the level of exposure to the cytotoxin released by enfortumab vedotin[78]. As we will discuss later, polymorphisms in genes encoding enzymes involved in the metabolism of ADCs and/or their payloads can result in increased toxicities as observed in patients with deleterious polymorphisms in *UGT1A1* who received sacituzumab govitecan[79,80]. Body weight and composition can both alter the metabolism of ADCs, potentially producing variability in toxicity profiles. Changes in body weight and changes in albumin concentration have both been consistently shown to affect the pharmacokinetics of T-DM1, T-DXd, tisotumab vedotin, enfortumab vedotin, mirvetuximab soravtansine and several other ADCs[78,81–84]. Lastly, ethnicity has been found to affect ADC metabolism: for example, patients of Japanese ethnicity have a 20% increase in mean serum T-DXd concentration compared to patients from other countries[81], a finding that might explain the higher incidences of ILD observed in this patient population[77].

Adverse effects associated with combination strategies

On the basis of their mechanism of action, ADCs offer multiple opportunities for combination strategies, with the aim of achieving either additive or synergistic antitumour activity. Together with the potential to improve clinical activity, however, these combinations also harbour the risk of increasing the toxicity of the regimen, either owing to overlapping adverse effects or unexpected synergies. In this section, we recapitulate the available toxicity data on ADC combinations administered alongside different classes of anticancer agents (Tables 3 and 4).

Table 3 | Toxicities observed with ADCs combined with chemotherapy in patients with solid tumours

Study details	Disease	Most common any-grade toxicities[a]	Most common grade ≥3 toxicities[a]	Notes
NCT00934856 (ref. 87), T-DM1+docetaxel±pertuzumab, phase Ib/IIa	$HER2^+$ advanced-stage breast cancer	MBC: neutropenia (76%), asthenia (72%), thrombocytopenia (64%), epistaxis (56%); similar toxicities in patients with LABC	MBC: neutropenia (72%), leukopenia (44%) LABC: neutropenia (29%), serum ALT elevations (15%), thrombocytopenia (12%)	Dose modifications owing to AEs required in 48% of patients (12/25), mostly for neutropenia; 5 patients had skin-related clinically serious AEs
NCT01702558 (ref. 88), T-DM1±capecitabine, phase I/II	$HER2^+$ metastatic breast cancer	Thrombocytopenia (43%), serum AST elevations (33%), nausea (33%)	Thrombocytopenia (10%), serum AST elevations (5%), increased γ-glutamyltransferase (5%)	Higher rate of treatment discontinuations owing to AEs with T-DM1+capecitabine vs T-DM1 alone (28% vs 15%)
NCT00951665 (ref. 193), T-DM1+paclitaxel+pertuzumab, phase Ib/IIa	$HER2^+$ metastatic breast cancer	Peripheral neuropathy (91%), fatigue (80%), nausea (45%), dry eye (41%), alopecia (38%)	Peripheral neuropathy (23%), neutropenia (20%), fatigue (18%)	Most patients (73%) discontinued paclitaxel owing to AEs; 2 treatment-related deaths reported
DESTINY-Gastric03 (ref. 89), T-DXd+5-FU or capecitabine, phase Ib/II	$HER2^+$ metastatic gastric or gastro-oesophageal cancer	NR	5-FU arm: anaemia (33%), neutropenia (33%), nausea (13%), stomatitis (13%) Capecitabine arm: neutropenia (40%), anaemia (30%), nausea (20%)	2 DLTs (grade 3 stomatitis) observed with T-DXd 6.4 mg/kg+5-FU 800 mg/m^2
TROPION-Lung02 (ref. 90), Dato-DXd+platinum chemotherapy+pembrolizumab, phase Ib	Advanced-stage NSCLC without actionable oncogenic alterations	Nausea (44%), decreased appetite (38%), fatigue (31%), anaemia (31%), vomiting (25%), stomatitis (23%)	Thrombocytopenia (13%), neutropenia (13%), stomatitis (6%), anaemia (6%), fatigue (6%)	Clinically serious AEs in 27% of patients, one patient had grade 3 ILD
NCT02606305 (ref. 91), mirvetuximab soravtansine+carboplatin, phase Ib	Platinum-sensitive ovarian cancer	Nausea (66%), diarrhoea (61%), thrombocytopenia (61%), blurred vision (61%), fatigue (55%), neutropenia (55%), vomiting (50%), hypokalaemia (44%), peripheral neuropathy (44%)	Neutropenia (28%), thrombocytopenia (17%), hypokalaemia (11%), anaemia (11%)	Grade 1–2 pneumonitis observed in 5 patients
NCT02751918 (ref. 194), anetumab ravtansine+pegylated-liposomal doxorubicin, phase Ib	Platinum-resistant ovarian cancer	Nausea (48%), decreased appetite (43%), fatigue (39%), diarrhoea (32%), corneal disorders (29%)	Fatigue (12%), neutropenia (11%), anaemia (9%)	1 death reported as treatment related (neutropenic sepsis)

5-FU, 5-fluorouracil; ADCs, antibody–drug conjugates; AEs, adverse events; ALT, alanine aminotransferase; AST, aspartate aminotransferase; Dato-DXd, datopotamab deruxtecan; DLTs, dose-limiting toxicities; ILD, interstitial lung disease; LABC, locally advanced breast cancer; MBC, metastatic breast cancer; NSCLC, non-small-cell lung cancer; NR, not reported; T-DM1, trastuzumab ematansine; T-DXd, trastuzumab deruxtecan.

Table 4 | Toxicities observed with ADCs combined with other therapies in patients with solid tumours

Study details	Disease	Most common any-grade toxicities	Most common grade ≥3 toxicities	Notes
ADCs plus endocrine therapy				
TALENT-TRIO[94], T-DXd ± anastrozole, phase II	$HR^+/HER2^{low}$ early stage breast cancer	Nausea (~80%), fatigue (~60%), vomiting (~50%), alopecia (~50%), diarrhoea (~40%), constipation (~40%)	Fatigue (5–10%), hypokalaemia (10%), diarrhoea (5–10%)	1 death owing to myocardial infarction, reported as possibly drug related; discontinuations owing to AEs in 3 patients
DESTINY-Breast08 (ref. 95), T-DXd + anastrozole or fulvestrant, phase Ib	$HER2^{low}$ metastatic breast cancer	NR	Anaemia (17–33%), thrombocytopenia (0–17%), hypokalaemia (17%), one patient had femoral neck fracture	1 patient in the T-DXd+fulvestrant arm had ILD
KATHERINE[195], T-DM1 + ET, phase III	$HER2^+$ early stage breast cancer with residual disease after NACT	Fatigue (48%), nausea (41%), headache (28%), increased AST (28%), arthralgia (26%), reduced platelet count (25%), radiation-related skin injury (24%), increased ALT (23%)	Decreased platelet count (4%), radiation-related skin injury (1%), peripheral sensory neuropathy (1%), fatigue (1%)	More patients discontinued treatment with T-DM1 alone than with T-DM1+ET (23% vs 16%)
ADAPT-TP[196], T-DM1 + ET, phase II	HR^+, $HER2^+$ early stage breast cancer	Fatigue (23%), nausea (21%), AST or ALT increases (19% each), headache (16%), thrombocytopenia (11%)	Grade 3 AEs observed in 8% of patients	Dose reductions were more frequent with T-DM1 alone (8%) than with T-DM1 plus ET (22%)
ADCs plus immunotherapy				
KATE2 (ref. 102), T-DM1 ± atezolixumab, phase III	$HER2^+$ metastatic breast cancer	Fatigue (38%), nausea (37%), pyrexia (35%), headache (27%), diarrhoea (23%), appetite loss (22%), rash (21%)	Thrombocytopenia (13%), increased AST (6%), anaemia (5%), neutropenia (5%), increased ALT (5%); all with an increased incidence compared with the T-DM1 monotherapy arm	More clinically serious AEs in the combination arm (33%) vs the monotherapy arm (19%); 1 fatal AE reported as treatment related with T-DM1+atezolizumab (haemophagocytic syndrome)
DS8201-A-U105 (ref. 104), T-DXd + nivolumab, phase Ib	$HER2^+$ or $HER2^{low}$ metastatic breast cancer	NR; 100% of patients had any-grade AEs	NR; 50% of patients had grade ≥3 AEs	Clinically serious AEs in 38% of patients; 1 patient died of ILD, reported as treatment related
DS8201-A-U105 (ref. 105), T-DXd + nivolumab, phase Ib	$HER2^+$ or $HER2^{low}$ metastatic urothelial carcinoma	Nausea (74%), fatigue (53%), vomiting (44.1%)	NR; 74% of patients had grade ≥3 AEs	ILD reported as drug related in 24% of patients, including 1 death
BEGONIA[106], T-DXd + durvalumab, phase Ib	$HER2^{low}$ metastatic TNBC	Nausea (77%), fatigue (51%), neutropenia (31%), vomiting (29%), alopecia (27%), reduced appetite (26%)	Neutropenia (22%), anaemia (8%)	8 patients (14%) had treatment-related ILD, including 1 death
BEGONIA[108], datopotamab deruxtecan + durvalumab, phase Ib	$HER2^{low}$ metastatic TNBC	Nausea (57%), stomatitis (56%), alopecia (46%), fatigue (39%), constipation (39%), rash (28%), vomiting (21%)	NR; 41% of patients had grade ≥3 AEs	3% of patients had ILD, no treatment-related deaths
EV-302 (ref. 197), enfortumab vedotin + pembrolizumab, phase Ib/II	Metastatic urothelial carcinoma	Peripheral neuropathy (56%), fatigue (51%), alopecia (49%), diarrhoea (47%), appetite loss (40%), rash (36%), dysgeusia (33%), pruritus (33%)	Peripheral neuropathy (4%), fatigue (11%), diarrhoea (4%), appetite loss (2%), rash (11%)	Clinically serious AEs in 16% of patients; 1 death reported as treatment related owing to multiple organ dysfunction syndrome
ADCs plus targeted therapy				
NCT02657343 (ref. 111), T-DM1 + ribociclib, phase Ib/II	$HER2^+$ metastatic breast cancer	Thrombocytopenia (92%), anaemia (83%), neutropenia (75%), fatigue (58%), AST increase (50%), ALT increase (43%), QT prolongation (42%), nausea (41%), hyperbilirubinaemia (25%)	Neutropenia (42%), leukopenia (33%), anaemia (25%), lymphopenia (17%), thrombocytopenia (17%), febrile neutropenia (8%), elevated AST or ALT (8%), hypophosphataemia (8%), urinary tract infection (8%)	Dosage schedule of ribociclib was changed from days 5–18 to days 8–21 of a 21-day cycle owing to thrombocytopenia in the setting of concurrent T-DM1 therapy

Table 4 (continued) | Toxicities observed with ADCs combined with other therapies in patients with solid tumours

Study details	Disease	Most common any-grade toxicities	Most common grade ≥3 toxicities	Notes
ADCs plus endocrine therapy (continued)				
NCT01983501 (ref. 110), T-DM1 + tucatinib, phase Ib	HER2$^+$ metastatic breast cancer	Nausea (72%), diarrhoea (60%), fatigue (56%), epistaxis (44%), headache (44%), thrombocytopenia (42%), vomiting (42%), constipation (42%), decreased appetite (40%)	Thrombocytopenia (28%), hypokalaemia (16%), ALT increase (14%), AST increase (14%), fatigue (12%)	Clinically serious AEs occurred in 37% of patients, including 2 cardiac failures
FORWARD II[113], mirvetuximab soravtansine + bevacizumab, phase Ib	FRα$^+$ epithelial ovarian carcinoma	Diarrhoea (52%), blurred vision (50%), nausea (45%), fatigue (41%), peripheral neuropathy (35%), thrombocytopenia (30%), dry eye (27%), hypertension (26%)	Hypertension (14%), thrombocytopenia (5%), ALT increase (5%)	Clinically serious AEs occurred in 42% of patients; 6 patients (9%) had grade 1–2 pneumonitis reported as treatment related

ADCs, antibody–drug conjugates; AEs, adverse events; ALT, alanine aminotransferase; AST, aspartate aminotransferase; ET, endocrine therapy; FRα, folate receptor-α; HR, hormone receptor; ILD, interstitial lung disease; NACT, neoadjuvant chemotherapy; T-DM1, trastuzumab ematansine; T-DXd, trastuzumab deruxtecan; TNBC, triple-negative breast cancer.

ADCs combined with chemotherapy

Combining different types of chemotherapies is an established method of overcoming resistance and achieving improved treatment outcomes[85]. Moreover, certain chemotherapies have been suggested to synergize with ADCs: gemcitabine, for example, has been shown to upregulate HER2 expression on cancer cells in vitro via activation of NF-κB signalling, thus enhancing the activity of T-DM1 (ref. 86). Combining ADCs with chemotherapy, however, comes with certain challenges related to overlapping toxicities.

Some of the data in this context originate from a trial testing T-DM1 in combination with docetaxel (with or without pertuzumab) in patients with HER2-positive breast cancer. Among patients with metastatic breast cancer, this combination led to several dose-limiting toxicities (DLTs) and grade ≥3 adverse events in around 80% of patients, with more than half of all patients having neutropenia, fatigue, epistaxis, stomatitis, nausea and diarrhoea[87]. T-DM1 has also been combined with capecitabine in a randomized phase II trial enrolling patients with HER2-positive metastatic breast cancer, demonstrating no improvement in response rate and more than doubling in the percentage of patients discontinuing treatment when adding capecitabine to T-DM1 monotherapy (from 16.3% to 35.8%)[88]. T-DXd was combined with 5-fluorouracil (5-FU) or capecitabine in the DESTINY-Gastric03 trial that enrolled 25 patients with metastatic HER2-positive gastric cancer. Dose-limiting stomatitis was observed in two patients in addition to a high incidence of grade ≥3 adverse events in the 5-FU combination arm (in 93.3% of patients), with most patients requiring reductions in the 5-FU dose; no DLTs were observed in the capecitabine arm, although the incidence of grade ≥3 adverse effects was high (90%), and 80% of patients required capecitabine dose reductions[89]. Dato-DXd was tested in combination with platinum-based chemotherapy and pembrolizumab, resulting in grade ≥3 toxicities in 60% of patients, with nausea, anaemia, fatigue and stomatitis all commonly observed[90]. Finally, mirvetuximab soravtansine was combined with carboplatin in a phase Ib trial, resulting in a high incidence of any-grade nausea (in 67% of patients) as well as any-grade vomiting, diarrhoea, ocular toxicities, fatigue and cytopenias, all in ≥50% of patients[91].

In summary, data from most trials seem to suggest a non-negligible increase in toxicities when combining ADCs with conventional chemotherapy, likely owing to overlapping toxicities related to the off-target, off-tumour effects of ADC payloads.

ADCs combined with endocrine therapy

Endocrine therapy is a common treatment strategy with the general objective of inhibiting the growth of hormone-dependent cancers by either blocking hormone production or the ability of hormones to promote tumour cell growth. Such agents are generally well tolerated and are most frequently (but not exclusively) administered to patients with either breast or prostate cancer[92]. The non-overlapping toxicity profiles of most endocrine therapies has enabled them to be combined with chemotherapy, and more recently with ADCs. In the phase III KATHERINE trial, which compared adjuvant T-DM1 to trastuzumab in patients with HER2-positive breast cancer with residual disease after neoadjuvant HER2-directed therapy, concomitant adjuvant endocrine therapy was permitted in both arms. The incidence of any-grade toxicities was similar among patients receiving T-DM1 with versus without endocrine therapy, as was the incidence of grade ≥3 adverse events (26.0% versus 24.9%), serious adverse events (12.9% versus 12.2%) and adverse events leading to T-DM1 dose reduction (11.0% versus 15.0%)[93]. Endocrine therapy has also been tested in combination with T-DXd, both in patients with early stage and advanced-stage HER2-low breast cancers. The randomized phase II TALENT trial, in which patients with early stage HER2-low breast cancer received neoadjuvant T-DXd either with or without anastrozole, demonstrated a comparable toxicity profile in both arms[94]. Similarly, in the phase Ib DESTINY-Breast08 trial, the addition of anastrozole or fulvestrant to T-DXd did not lead to DLTs and was associated with a toxicity profile similar to that of T-DXd alone in patients with metastatic HER2-low breast cancer[95]. Overall, combining ADCs with endocrine therapy does not seem to be associated with an increase in toxicities, consistent with the different adverse effect profiles of each agent alone as well as the favourable toxicity profile of most endocrine therapies relative to other systemic anticancer therapies.

ADCs combined with immunotherapy

ADCs share with chemotherapy the potential to elicit immunogenic cell death[96–98] while also harbouring a potentially immune-stimulating function through the Fc domain of the antibody[99], thus providing a rationale for combination with immune-checkpoint inhibitors (ICIs)[100]. This rationale is supported by the observation that the adverse effect profiles of ADCs and ICIs are generally non-overlapping, suggesting that combination therapy is feasible[101]. Several early phase and late-phase

trials have tested various combinations of ICIs and ADCs, including at least one large randomized study. The phase III KATE2 trial tested the addition of atezolizumab to T-DM1 in patients with pretreated HER2-positive metastatic breast cancer, demonstrating a moderate increase in toxicities in the combination arm: one treatment-related death occurred in a patient receiving T-DM1 plus atezolizumab, and the incidence of clinically serious adverse events (33% versus 19%) as well as that of most adverse effects, with particular mention of pyrexia (35% versus 16%, including several cases that led to hospitalization), were all increased with the addition of atezolizumab[102]. Randomized data are also available for enfortumab vedotin with versus without pembrolizumab in a study involving 149 patients with advanced-stage urothelial carcinoma: the addition of pembrolizumab led to an increase in the incidence of clinically serious treatment-related adverse events (23.7% versus 15.1%), fatal treatment-related adverse events (3.9% versus 2.7%) and a general increase in the incidence of all treatment-related adverse events, with particular mention to severe skin reactions[103]. Nonetheless, this combination received accelerated FDA approval in April 2023 for patients with locally advanced or metastatic urothelial carcinoma who are ineligible for cisplatin-containing chemotherapy based on the compelling efficacy of this combination. The non-randomized design of most other studies testing ADCs plus ICIs precludes definitive answers to many of the outstanding research questions. Nonetheless, no concerning signals of synergistic toxicity have been observed thus far with the combination of ICIs and T-DXd[8,104,105], Dato-DXd[106] or sacituzumab govitecan[107], all of which have a mostly additive toxicity profile. These safety profiles include no notable differences in both the rate and severity of ILD observed with DXd-based ADCs, which do not seem to increase following the addition of ICIs[104,106,108,109]. Further data in this space are expected from ongoing randomized phase III trials (NCT05629585, NCT05382286 and NCT05633654), which are expected to provide clarification on the toxicity profiles of regimens combining ICIs with ADCs beyond T-DM1.

ADCs combined with targeted therapies

Among currently approved ADCs, T-DM1 is the agent with the largest amount of evidence available on activity and safety when administered in combination with targeted agents. The combination of T-DM1 with the HER2 tyrosine kinase inhibitor tucatinib was tested in a phase Ib trial and was found to be tolerable, albeit with frequent gastrointestinal and hepatic toxicities, with 37% of the patients having at least one clinically serious adverse event and 56% of patients requiring tucatinib dose interruptions[110]. Data from the ongoing phase III HER2CLIMB-02 trial testing T-DM1 with versus without tucatinib in patients with locally advanced or metastatic HER2-positive breast cancer are expected to be reported soon, and these might clarify the added toxicity of the combination over that of T-DM1 alone. T-DM1 has also been tested in combination with intermittent CDK4 and CDK6 inhibition using ribociclib in a phase Ib trial, with no DLTs observed, although 58% of patients required ribociclib dose reductions owing to thrombocytopenia or neutropenia and one patient had grade 2 QTcF prolongation[111]. Sacituzumab govitecan was combined with the PARP inhibitor talazoparib in a phase Ib trial enrolling patients with metastatic TNBC, leading to multiple DLTs owing to severe myelosuppression described in the first study report (most of the enrolled patients had febrile neutropenia)[112]. Lastly, the addition of the anti-VEGFA antibody bevacizumab to mirvetuximab soravtansine was evaluated in a phase Ib trial involving patients with platinum-resistant ovarian cancer[113], resulting in a toxicity profile comparable to that of the ADC alone[114], although grade 1–2 pneumonitis was observed with the addition of bevacizumab in six (9%) patients versus none with mirvetuximab soravtansine alone.

Emerging strategies to optimize the safety of ADCs

In the short time frame of 4 years, the number of ADCs approved as treatments for patients with cancer has doubled, resulting in a rapidly expanding population of patients exposed to these agents. This rapid pace of development warrants extensive efforts in the development of practices to minimize the risks associated with receiving ADCs. Several strategies have been pursued to either prevent or optimally manage the toxicities associated with ADCs in clinical practice (Fig. 3).

Dose-optimization strategies

Given the dose dependency of many of the toxicities associated with ADCs, considerable attention has been dedicated to the optimization of both dose and administration schedules, in an attempt to improve the therapeutic index[115]. Five classic dose-optimization strategies have been adopted for this purpose, including body weight dose capping, treatment duration capping, optimization of dose frequency, response-guided dose adaptation and randomized dose-finding studies.

Dose capping. The rationale for dose capping derives from the influence of body weight on the pharmacokinetic profile of ADCs, which can lead to risks of overdosing for patients with greater body weight. A dose cap of 125 mg was adopted for the treatment of patients with urothelial carcinoma with enfortumab vedotin after three fatal adverse events reported to be related to treatment were observed with this ADC in patients with baseline body weights ≥100 kg (ref. 116). Analogously, dosing based on adjusted ideal body weight was adopted for mirvetuximab soravtansine in order to reduce the incidence of ocular adverse events[117]. In general, dose capping should always be considered when, during the development of an ADC, concerning adverse effects are observed among patients with body weight above the ideal range and/or pharmacokinetic modelling suggests a relevant effect of patient body weight on drug exposure.

Capping of treatment duration. Capping of treatment duration aims to minimize the incidence of chronic and potentially permanent adverse events such as peripheral neuropathy. This approach is well established for unconjugated chemotherapies, such as taxanes[118] or platinum salts[119], and has also proven beneficial with ADCs. For example, data from parametric time-to-event analyses indicate that the predicted risk of developing grade ≥2 peripheral neuropathy with polatuzumab vedotin increases by ≥50% when eight versus six treatment cycles are administered[120]. These data supported the approval of polatuzumab vedotin for patients with relapsed and/or refractory diffuse large B cell lymphoma at a dose of 1.8 mg/kg every 3 weeks for a maximum of six cycles[121]. This approach has not yet been extensively tested in patients with solid tumours but has the potential to offer improved tolerability and safety across several different ADCs and cancer types.

Fractionated administration. The frequency of administration is another crucial variable that, when modulated, might improve the safety of ADCs. Administration of ADCs using fractionated dosing schedules can result in exposure to the same cumulative dose of the drug albeit with a lower peak plasma concentration (C_{max}) compared to that achieved with a single but higher dose, potentially ameliorating adverse events driven by the C_{max}. The first successful example of this

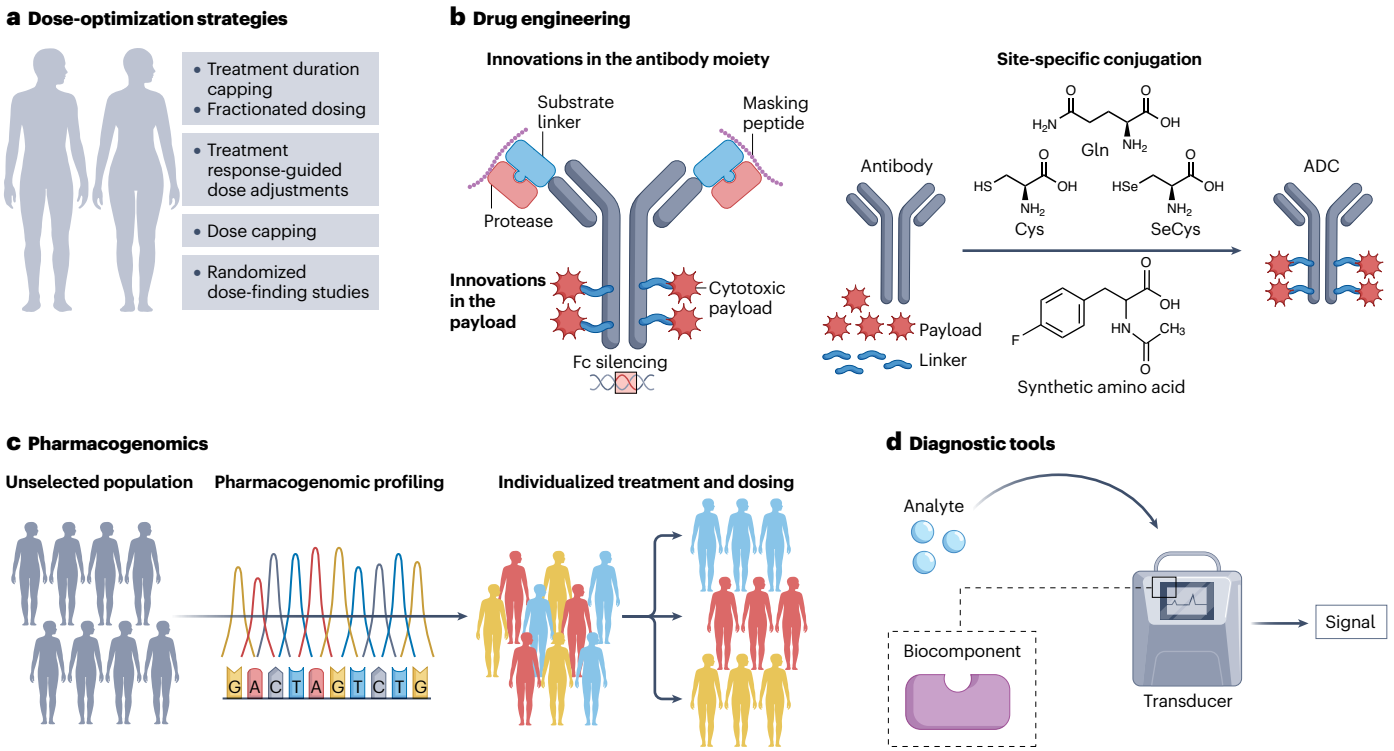


Fig. 3 | Strategies being tested to potentially optimize the safety of ADCs. Multiple strategies designed to improve the safety profile of antibody–drug conjugates (ADCs) and thus increase their therapeutic index are currently being developed. **a**, Dose-optimization strategies such as treatment duration modifications and fractionated dosing, treatment-response dose adjustments, dose capping, and designing randomized dose-finding studies. **b**, Drug-engineering approaches, including but not limited to the generation of probody–drug conjugates, silencing the Fc portion of the ADC to avoid Fc receptor-mediated toxicities and novel site-specific conjugation technologies. **c**, Using pharmacogenomic profiling to inform on the risks of developing adverse events in each individual patient and tailoring ADC administration accordingly. **d**, Using wearable biosensors to detect and monitor specific toxicities associated with ADCs.

strategy was achieved with the first-ever approved ADC, the CD33-targeted agent gemtuzumab ozogamicin. This compound was first granted accelerated approval at the dose of 9 mg/m^2 every 2 weeks for patients with relapsed CD33-positive acute myeloid leukaemia in 2000 (ref. 122). However, a high incidence of liver toxicity and veno-occlusive disease was reported in real-world studies, leading to its withdrawal from the market in 2010 (ref. 122). On the basis of subsequent evidence of improved safety, gemtuzumab ozogamicin was reapproved in 2017 with a fractionated, reduced-dose dosing schedule (days 1, 4, and 7 of each induction cycle)[2,122]. Another example of a successful dose frequency adjustment that improved ADC safety is provided by the CD22-targeted ADC inotuzumab ozogamicin, which was found to be more tolerable and equally effective when administered at fractionated weekly rather than three-weekly or four-weekly doses, leading to the current approval for patients with acute lymphoblastic leukaemia at an induction dose of 1.8 mg/m^2 administered as three-weekly doses, with recommended dosing for subsequent cycles dependent on the initial response[123–125]. Similar approaches with ADCs currently used to treat patients with solid tumours might be warranted, particularly given the uncertain implications of C_{max} for the activity of most ADCs[115].

Treatment response-guided dose adjustments. Response-guided dose adjustments provide a method of modulating the extent of exposure to an ADC based on the initial response of each patient. This adaptive dosing strategy is currently used for patients receiving inotuzumab ozogamicin, for which the initial dose of 1.8 mg/m^2 is reduced to 1.6 mg/m^2 in subsequent cycles in patients who enter complete remission in order to reduce the risk of toxicities[123]. Testing this strategy in patients with solid tumours seems most suitable for indications in which responses to the ADC are frequent and long remissions can be expected such as patients with HER2-positive metastatic breast cancer receiving T-DXd. In the phase III DESTINY-Breast03 trial involving patients with trastuzumab-refractory HER2-positive metastatic breast cancer, treatment with T-DXd resulted in complete remissions in 21% of patients, with more than half of all patients in the T-DXd arm being free of progression at 2 years[126]. Given the known association between a complete response and long-lasting remission observed in this disease[127], the evaluation of response-guided T-DXd dosing might be reasonable, and similar attempts could be pursued with other ADCs.

Randomized dose-finding studies. Randomized dose-finding studies are prospective trials designed to evaluate more than one dosage of a compound, with the purpose of identifying an optimal dose that maximizes the therapeutic index. Randomized dose-finding studies enabled the optimal dose of T-DXd to be identified for patients with HER2-positive breast cancer and those with *HER2*-mutant non-small-cell lung cancer (NSCLC): for both indications, randomized trials

were conducted to compare the benefit-to-risk profile of 5.4 mg/kg and 6.4 mg/kg administered as three-weekly doses. In the setting of HER2-positive breast cancer, 115 patients were randomized to receive T-DXd at either 5.4 mg/kg or 6.4 mg/kg, and each group had a similar objective response rate (ORR; 56.5% and 61.5%, respectively), albeit with 12% and 21% of patients developing ILD, and a substantial increase in the incidence of grade ≥3 treatment-emergent adverse events at the higher dose (39% and 58%, respectively)[128]. With regard to *HER2*-mutant metastatic NSCLC, 80 patients were randomized 2:1 to receive T-DXd at doses of either 5.4 mg/kg or 6.4 mg/kg, and both groups had similar ORRs (53.8% and 57.7%, respectively), albeit with a near doubling of the incidence of grade ≥3 adverse events (from 31.7% to 58.0%), including an increased incidence of ILD with the higher dose (5.9% at 5.4 mg/kg versus 14% at 6.4 mg/kg)[129]. T-DXd was approved for patients with *HER2*-mutant metastatic NSCLC at the dose of 5.4 mg/kg based on these data. Receiving a higher dose of T-DXd was confirmed to be associated with a higher risk of ILD in a pooled analysis of data from 1,150 patients receiving the drug, of whom 44% had breast cancer and the remaining 56% had solid tumours of other histologies[77]. Similar attempts at randomized dose-finding studies might be reasonable for other ADCs and indications, particularly for those in which fatal adverse events and/or considerable grade ≥3 toxicities are observed either in trials or clinical practice.

Optimizing ADC design

ADC engineering and other optimization strategies can provide important methods of maximizing both the efficacy and safety of these agents. Indeed, innovations in the design of each ADC component can enable fine-tuning of the pharmacological properties, with potential implications for tolerability.

Innovations in the antibody moiety. Most of the currently approved and investigational ADCs are directed against targets that are, to varying degrees, heterogeneously expressed in non-malignant tissues. The generation of probody–drug conjugates (PDCs), consisting of conditionally activated ADCs in which the antigen-binding regions of the antibody are masked by protease-cleavable peptides that enable selective unmasking and activation within the TME, provides one example of an engineering strategy that might reduce the incidence of on-target, off-tumour toxicities[130]. PDCs are indeed expected to remain largely intact in non-malignant tissues while being effectively cleaved by tumour-associated proteases in malignant tissues in which the masking peptide is released, enabling the antibody to bind the target antigen and the payload to reach tumour cells[130]. This approach has the additional advantage of potentially overcoming the binding site barrier seen in most solid tumours owing to delayed antigen binding and improved tumour penetration, which could lead to improved efficacy and an improved therapeutic index[131,132]. Of note, early clinical data with PDCs have yet to reveal clinically relevant improvements in the safety profile as observed in a phase I/II study in which patients with advanced-stage solid tumours received the CD166-targeted PDC pralu-zatamab ravtansine: ocular toxicities occurred in 43% of patients and 37% of patients had ≥1 treatment-emergent grade ≥3 adverse event[14,133]. Nonetheless, multiple other PDCs are in development, with additional data required to draw conclusions on the clinical utility of this technology. Beyond masking of the binding regions of the antibody, attempts to silence the antibody Fc domain with the aim of reducing off-target, off-tumour toxicities related to the Fc-mediated uptake of the ADC by immune cells are currently ongoing[134,135].

A different strategy that might enable improved ADC delivery to the tumour site and potentially reduce the incidence of on-target toxicity involves conjugation of the payload to bispecific rather than conventional antibodies. By targeting two different antigens, bispecific antibodies might have increased selectivity and improved internalization in tumour cells compared with conventional antibodies[136–139]. The first clinical data on the safety and activity of a bispecific ADC were presented in September 2022, with the HER2-targeting ADC ZW49 (directed against two non-overlapping HER2 epitopes) demonstrating an ORR of 31% in patients with advanced-stage HER2-expressing solid tumours, albeit with a non-negligible toxicity profile (mostly owing to ocular toxicities, including keratitis in 42%)[140]. Several other bispecific ADCs are currently being developed, including, among others, those targeting different HER2 epitopes, HER2 and Trop2, HER2 and PRLR, HER2 and CD63, EGFR and MUC1, and EGFR and HER3 (refs. 141–145).

Innovations in linker technology. Multiple strategies designed to improve the stability of ADCs in systemic circulation are currently being developed in an attempt to optimize safety. Most of these efforts focus on the site-specific rather than stochastic conjugation of payloads to antibodies, via either engineered cysteines[146–148], the introduction of unnatural amino acids[149,150], and/or the addition of selenocysteine, glutamine or aldehyde tags[151]. These methods enable the preparation of homogenous populations of ADCs with predictable DARs and sites of payload attachment, which might be associated with an improved tolerability profile and more predictable pharmacokinetics. Preclinical evidence suggests that these conjugation strategies confer improved safety profiles, and clinical experimentation is currently ongoing[56]. Nonetheless, as previously mentioned, clinical data suggest that even excess linker stability can be associated with severe toxicities as observed in the clinical trials of SYD985, ARX788 and A166, among other ADCs[45,54,55]. A different engineering strategy involves adding short polyethylene glycol (PEG) moieties to the linker, which might reduce the incidence of non-specific ADC uptake via an increase in ADC hydrophilicity, ultimately reducing the incidence of off-target toxicities[152]. Preclinical data suggest retained activity and improved pharmacokinetic properties for ADCs developed using this strategy[153–155], and clinical testing is currently ongoing.

Innovations in the payload. In an attempt to improve the therapeutic index of single-payload ADCs, novel constructs featuring two different types of payload linked to the same monoclonal antibody have been developed. Together with potentially expanding the spectrum of activity of the ADC, the combination of payloads with multiple and non-overlapping mechanisms of action within the same conjugate might also lead to improved tolerability. Preclinical data have confirmed the feasibility and high antitumour activity of this approach with constructs such as a HER2-targeted antibody conjugated to both MMAE and MMAF and an FGF2-targeted antibody conjugated to MMAE and α-amanitin[156–159], although no clinical data are currently available. Furthermore, novel payloads, beyond anti-microtubule agents, topoisomerase I inhibitors and DNA-intercalating agents, are being explored. These payloads include other cytotoxic molecules (such as topoisomerase II inhibitors or agents that inhibit transcription or translation)[48] as well as ADCs with unconventional payloads such as immunostimulatory molecules, heterobifunctional protein degraders and tyrosine kinase inhibitors[48]. A detailed description of each of these strategies is beyond the scope of this Review; however, it is worth highlighting that non-conventional payloads might be subject

to toxicity issues similar to those observed with canonical cytotoxin-conjugated ADCs and might warrant careful experimentation and fine-tuning. As an illustration, preliminary data from a phase I study testing a HER2-targeted, immune-stimulating ADC carrying a Toll-like receptor 7 agonist were reported in December 2022 and revealed high incidences of cytokine-release syndrome, pyrexia, nausea, vomiting and headache (>30% each), consistent with the off-target effects that might be expected from release of the immunostimulatory payload[160]. Moreover, the development of the immune-stimulating STING-agonist ADC XMT-2056 was halted in March 2023 after a fatal adverse effect deemed to be related to the conjugate was observed in the second patient treated at the initial dose level in the dose-escalation portion of the trial[161]. In these settings, leveraging the knowledge accumulated through the development of conventional ADCs is warranted in order to develop safer innovative ADCs that are able to effectively expand the range of therapeutic options.

The co-administration of ADCs with payload-neutralizing antibody fragments is another innovation designed to improve the tolerability of ADCs. By binding to unconjugated circulating payload molecules, these payload-neutralizing fragments are designed to reduce the amount of cytotoxin reaching non-malignant tissues and thus reduce the extent of off-target toxicities in patients receiving ADCs. Preclinical data indicate a reduction in chemotherapy-related toxicities, amelioration of body weight loss in mouse models and retained activity when a HER2-targeted MMAE-based ADC is co-administered with the humanized anti-MMAE Fab fragment ABC3315 in xenograft models[162]. Clinical investigations will be required to clarify whether this strategy can effectively improve tolerability without affecting the antitumour activity of ADCs.

Pharmacogenomics

The identification of patients with the greatest risk of developing adverse events upon treatment with ADCs is an important step in improving the clinical management of such patients. In a post hoc analysis of data from the ASCENT trial, in which patients with metastatic TNBC received sacituzumab govitecan, deleterious *UGT1A1* polymorphisms were noted to correlate with the occurrence of adverse events reported to be related to treatment[80]. The payload of sacituzumab govitecan (SN-38) is a lipophilic molecule that is largely cleared via glucuronidation by UGT1A1. Data from previous studies indicate that patients with colorectal cancer receiving irinotecan who have loss-of-function *UGT1A1* polymorphisms have increased exposure to SN-38, resulting in higher incidences of both neutropenia and diarrhoea[163,164]. This effect was also observed among patients homozygous for the *UGT1A1**28 polymorphism receiving the combination of capecitabine and irinotecan, who were found to be 14.2 times more likely to develop febrile neutropenia than those with wild-type *UGT1A1* (ref. 165). In ASCENT, the incidences of grade ≥3 neutropenia, febrile neutropenia and diarrhoea were all higher among patients who were homozygous for *UGT1A1**28 compared with patients who were either heterozygous for this allele or were homozygous for the wild-type allele[80]. Similarly, patients who were homozygous for *UGT1A1**28 had an increased incidence of toxicities on sacituzumab govitecan in the phase I/II IMMU-132-01 basket trial[79]. Given the relatively low frequency of the homozygous *UGT1A1**28 genotype (<10%)[163], these data have not yet led to recommendation for routine testing of *UGT1A1* polymorphisms, although monitoring patients who are known to be homozygous for *UGT1A1**28 more closely for the development of toxicities is currently recommended. In addition to polymorphisms in genes encoding enzymes involved in payload metabolism, polymorphisms affecting genes that encode enzymes involved in payload release or in those encoding Fc receptors might also have a role in determining the safety profile and tolerability of a given ADC in specific patient subgroups. Overall, these data suggest that pharmacogenomic parameters can have a relevant effect on the safety profile of certain ADCs. As novel ADCs are being developed, considering the inclusion of pharmacogenomic profiling when designing early phase trials could be a reasonable approach that ensures that the safety profile is not overly affected in specific populations and/or by inter-individual genetic variations.

Diagnostic tools

Another potential approach for the early detection and better management of ADC-induced adverse events involves the use of wearable biosensors (WBSs). These devices have the potential to provide real-time information through the continuous, non-invasive measurement of biological factors, including vital parameters and biofluids (such as sweat, saliva and interstitial fluid). In the context of ADC-related toxicities, the use of WBSs might support the early identification of treatment-induced ILD by detecting changes in oxygen saturation levels and/or heart and breathing rates. Moreover, forced vital capacity assessments using home spirometry (such as a Bluetooth-enabled hand-held spirometer), pulse oximetry or cough monitors (such as the Leicester Cough Monitor or VitaloJAK) have emerged as potential methods of monitoring for ILD[166–168]. The rapid development of these technologies might enable early diagnosis of ILD in patients at risk, help identify acute exacerbation of ILD and support real-time clinical decision-making, although several barriers to adoption of these measures also exist, including costs and the need for international, evidence-based guidelines regarding the use of home-based measurements.

ADCs in the early stage setting

ADCs have reshaped the treatment algorithms for the management of several forms of advanced-stage solid tumours over the past decade[13]. However, the greatest clinical benefit from this class of agents might be achieved in patients with early stage disease who often receive chemotherapy in an attempt to prevent disease recurrence and who might benefit from the improved delivery of cytotoxic agents provided by ADCs, ideally including cure for a higher percentage of patients. Several trials are currently ongoing in this space (Supplementary Table), with one ADC (T-DM1) already approved for patients with early stage HER2-positive breast cancer[169]. Importantly, this setting raises unique challenges in terms of toxicities: although increased incidences of short-term toxicities are commonly considered acceptable to achieve the goal of cure, permanent toxicities that might have long-term effects on quality of life are less acceptable and caution is needed to ensure that fatal toxicities can be avoided[170,171]. In this context, early stopping rules in trials testing ADCs in the early stage setting may be warranted.

The post-neoadjuvant setting, where ADCs can provide a rescue strategy for patients with residual disease at surgery (those without a pathological complete response) following preoperative systemic treatment is an appealing setting in which to test novel ADCs. The only ADC approved currently for this scenario is T-DM1, which was shown to improve disease-free survival in the phase III KATHERINE trial[169]. In this study, toxicities were more frequent with adjuvant T-DM1 compared with trastuzumab (grade ≥3 toxicities occurred in 25.7% versus 15.4% of patients) although no fatal toxicities were observed; peripheral neuropathy was also more frequent with T-DM1 (18.6% versus 6.9%),

which resolved in most patients during follow-up. Given the favourable risk-to-benefit profile, T-DM1 has rapidly become the standard-of-care adjuvant therapy for patients with residual invasive disease following neoadjuvant therapy and is recommended for this indication in most international guidelines[172,173]. Concomitantly, this success has raised the hope that novel ADCs could further improve outcomes in this high-risk setting. The ongoing phase III DESTINY-Breast05 trial (NCT04622319) is comparing post-neoadjuvant T-DXd to T-DM1 in a population similar to that of KATHERINE. Of note, T-DXd resulted in fatal ILD in 1–2% of patients in several studies conducted in the advanced-stage setting[77], a finding that indicates a need for extensive caution in the use of this agent in earlier settings. Sacituzumab govitecan is also being tested in the post-neoadjuvant setting within the phase III SASCIA trial (NCT04595565), in which patients with HER2-negative breast cancer and residual invasive disease after neoadjuvant chemotherapy are being randomized to either sacituzumab govitecan or treatment of physician choice. At the first interim safety analysis, patients receiving sacituzumab govitecan had an increased incidence of grade 3–4 adverse effects (66% versus 20% with treatment of physician choice) as well as more dose delays and treatment discontinuations, although no fatal toxicities were observed[174]. Analogously, the phase III OptimICE-RD (NCT05633654) and Tropion-Breast03 (NCT05629585) trials are currently testing sacituzumab govitecan and Dato-DXd in this setting, respectively.

ADCs are also under investigation in the preoperative setting. In the phase III KRISTINE trial, neoadjuvant T-DM1 plus pertuzumab failed to improve the outcomes of patients with HER2-positive breast cancer compared with chemotherapy and dual HER2 blockade with trastuzumab plus pertuzumab[175]. Neoadjuvant T-DXd (6–8 cycles) was evaluated in patients with HER2-low breast cancer in the phase II TALENT trial: most patients completed treatment and underwent surgery, although 12% came off study prior to treatment completion, either owing to death (one patient), withdrawal of consent (two patients) or meeting the discontinuation criteria[94]. The phase III DESTINY-Breast11 study (NCT05113251) is currently testing T-DXd as a substitute for either all or part of the traditional neoadjuvant chemotherapy regimen for HER2-positive breast cancer, whereas the phase II TRUDI trial (NCT05795101) is testing the combination of neoadjuvant T-DXd plus durvalumab for patients with non-metastatic, HER2-expressing inflammatory breast cancer. Neoadjuvant sacituzumab govitecan is being tested in patients with TNBC, with a first report coming from the randomized phase II NeoSTAR trial demonstrating a limited need for dose reductions (6%), no discontinuations owing to adverse events and no fatal toxicities[46]. Neoadjuvant enfortumab vedotin (three cycles) was evaluated for patients with urothelial carcinoma ineligible for cisplatin: 86% (19 of 22) of patients were able to complete treatment, with 18% having one or more grade 3–4 treatment-related adverse event and three fatal adverse events, although none of these were reported to be related to treatment[176].

Window-of-opportunity trials, a study design in which newly diagnosed patients receive a short course of treatment with a novel therapy before surgery or before initiating standard neoadjuvant treatment, usually with biological surrogates as end points, provide a novel method of testing potential neoadjuvant therapies, including ADCs[177]. Such trials might expedite drug development by providing evidence of the biological effects of the novel ADC in patients with treatment-naive tumours within a short time frame. For example, HER3-DXd was evaluated in the window-of-opportunity TOT-HER3 trial (NCT04610528), in which a single dose of HER3-DXd was administered before surgery to 78 patients with hormone receptor-positive breast cancer. This study demonstrated an increase in CelTIL score, which indicates an increase in tumour-infiltrating lymphocytes and/or a decrease in tumour cellularity, after treatment among responders as well as a lack of an association between baseline HER3 expression and treatment response[178]. These data informed the design of the ongoing randomized phase II SOLTI-VALENTINE trial (NCT05569811), which is testing the activity of six neoadjuvant cycles of HER3-DXd (with or without letrozole) or chemotherapy in patients with hormone receptor-positive breast cancer.

A final novel strategy involves the administration of ADCs to previously untreated patients in the adjuvant setting. This strategy was tested in the randomized phase II ATEMPT trial, in which 1 year of adjuvant T-DM1 led to a 5-year invasive disease-free survival of 97% among patients with resected stage I HER2-positive breast cancer[179]. One year of adjuvant T-DM1 was found to be safe, with an incidence of clinically relevant toxicities not significantly different to that observed with paclitaxel and trastuzumab (46% versus 47%; $P = 0.83$)[180]. Cardiac toxicities with adjuvant T-DM1 were also extremely rare in ATEMPT (in <1% of patients)[181]. A second study (ATEMPT 2.0, NCT04893109) is currently ongoing and is designed to determine whether a shorter course of T-DM1 (six cycles) can decrease the incidence of toxicities while retaining clinical activity.

Additional trials testing ADCs in the early stage setting are currently ongoing in patients with various tumour types (Supplementary Table). Nonetheless, the incidences of high-grade toxicities and treatment discontinuations observed with novel ADCs in this setting are non-negligible, indicating that a careful balance of potential risks versus benefits is warranted. This balance might be best optimized in the post-neoadjuvant setting, with the selective treatment of patients with residual disease after neoadjuvant therapy, a feature that has been associated with inferior outcomes across several cancer types[182–187].

Conclusions

ADCs have improved upon the activity of traditional chemotherapy regimens in a range of solid tumours. Despite their ideally targeted mechanism of action, most ADCs still confer frequent and sometimes life-threatening toxicities. Given the rapid expansion in their indications, an awareness of these adverse effects and their management is crucial as are efforts to prevent and mitigate ADC-related toxicities. These include careful attention to the risks of synergistic or overlapping toxicities when combining ADCs with other anticancer agents as well as caution in the development and testing of novel ADCs in early stage disease, in which a dedicated balance of risks and benefits is warranted for each indication. Innovations in ADC design, pharmacogenomic testing and WBSs as well as increased attention to the optimization of ADC dosing through dedicated prospective trials might help to leverage the potential of this highly promising class of anticancer drugs, which is still far from being fully explored.

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Author contributions

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Competing interests

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